Do Birds of a Father Flock Together? Examining Religious Homophily Dynamics at a Catholic University

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Abstract

In a context where the distribution of a salient dimension of association is skewed toward one group, are minorities more likely to band together (minority distinctiveness) or are majorities more likely to do so (majority exclusiveness)? We address this question by examining religious homophily and friendship choice using longitudinal network data from the *NetHealth* Cohort Study conducted at the University of Notre Dame ($N = 722$ students), a network ecology in which religious affiliation is a salient dimension of association. Using a dyadic logistic regression framework with an opportunity offset, we adjust for baseline group sizes and physical and social foci, such as residential proximity and socio-demographic homophily. We find that all religious groups are more likely to be selectively associated with co-religionists. Pooling tie-level data across six longitudinal waves with clustered standard errors, we find that non-Catholic religious affiliation, considered collectively (Protestant, No Religion, and Other Religion), has a stronger positive effect on active homophily compared to the Catholic majority, supporting Minority Distinctiveness Theory at the Catholic/non-Catholic level, and this pattern persists after adjusting for each ego's nomination activity and each alter's popularity in the friendship network, which suggests that it is not driven by a degree-based confound.

1 Introduction

1.1 Background

How does the numeric composition of a local social environment shape the formation of friendship ties across religious groups? While the reliable tendency for individuals to associate with similar others—homophily—is one of social science's most well-established empirical generalizations (McPherson et al., 2001), explaining *why* different groups exhibit varying degrees of homophily within a given population remains an open question (McFarland et al., 2014). Although selectively forming ties with similar others can be an individual choice, it is also influenced by structural factors such as propinquity, opportunity structure, and social foci (Rivera et al., 2010; Feld and Grofman, 2009).

In this last regard, network analysts distinguish between "choice homophily" resulting from individual preferences, proclivities, and relational incentives and "induced homophily" resulting from structural processes such as uneven group sizes or physical proximity, which makes partners of a certain kind more available than others (Kossinets and Watts, 2009; McPherson et al., 2001; Marsden, 1988; Hofstra et al., 2017). Structural clustering can also emerge indirectly via "consolidation" or multiform heterogeneity (Blau, 1977a), in which preferences for similarity along one dimension (e.g., gender or dormitory) narrow the pool of available alters, thereby altering apparent homophily along other dimensions.

In relation to the choice/structural homophily distinction, two major contemporary theories make somewhat divergent predictions about the effect of skewed group sizes on the strength of homophily tendencies. One theories hews more closely to the "agency" (choice homophily and relational incentives) side, and the other comes closer to the structural (strength in numbers) side.

Under the *Minority Distinctiveness Hypothesis* (MDH), a numeric minority status increases the cognitive salience of that group identity in the context of outgroup numerical dominance, which is theorized to produce active sorting patterns consistent with minority members banding together to preserve their subcultural boundaries (Mehra et al., 1998; Leonard et al., 2008; Marsden, 1988; Sepulvado et al., 2015).

Conversely, following Blau's classic treatment, what we might refer to as the *Majority Exclusiveness Hypothesis* (MEH) suggests that a dominant majority possesses the structural and social leverage to guard group boundaries, which means that they should exhibit stronger tendencies towards homophily, doubly aided by their standing as more numerous and more readily available tie-formation partners (majority members are more numerically available) and "induced" (there are more members of the majority to choose from) considerations. As Blau observed, "the larger of two groups discriminates more than the smaller against associating with members of the other group," resulting in higher active homophily among the majority (Blau, 1977a, p. 81).

1.2 Study Setting

An ideal setting for examining the MDH and MEH predictions would be a highly specific network ecology in which a single group holds overwhelming numerical dominance. The University of Notre Dame (ND) presents such a unique social laboratory for the present research (Hachen et al., 2022).

In the *NetHealth Study*'s (Purta et al., 2016) longitudinal cohort, Catholic students represent approximately 72.7% of the student body, while Protestant (9.8%), non-religious (12.3%), and other religious students (4.6%) constitute small, distinct minorities. However, this setting introduces a profound institutional focus (Feld, 1981; Feld and Grofman, 2009): religion is not merely a demographic attribute; it defines the university's very identity and physical infrastructure. Crucially, Catholics attending a Catholic university might be more motivated to befriend

other Catholics because the environment "summons" their religious identity, increasing its salience (Tavory, 2016). Furthermore, ND features built-in residential and social foci—such as single-sex dormitories, mandatory dorm assignments, and Catholic-oriented campus activities—that could organically channel Catholics into homophilous friendships without requiring any active individual preference for Catholic friends; our tie-formation data cannot itself establish whether such preferences are also present, only whether observed mixing patterns exceed what these structural mechanisms would predict on their own.

To evaluate the predictions of the MDH (higher homophily rates among minorities) versus the MEH (higher homophily rates among the majority), it is important to move beyond the limitations of raw homophily rates and aggregate indices. Traditional aggregate measures, such as Krackhardt's E-I (External-Internal) index (Krackhardt and Stern, 1988), suffer from a well-known artifact: in a bounded network, a numeric minority is structurally forced to have higher outgroup contact, regardless of their personal preferences, a phenomenon also emphasized by Blau (1977b), thus producing a form of "induced heterophily." Consequently, aggregate comparisons are biased (Everett and Borgatti, 2012). Here, we address this issue by transitioning to a dyadic (tie-level) framework with an opportunity offset and moving toward margin-free measures of group-level homophily, such as Yule's Q (Borgatti et al., 2024). This approach allows us to adjust for baseline group sizes in the data, as well as other important drivers of homophily, such as physical propinquity and social foci, and for other forms of homophily beyond the focal one in our study.

To this end, this paper leverages 8 waves of longitudinal network data from the *NetHealth Study* to examine the trajectory of religious homophily. We test whether the observed homophily of Protestants and other religious minorities is merely a byproduct of residential sorting and alternative tie-level demographic similarities, or whether it reflects a recurrent, active sorting process consistent with the identity-salience mechanism amid Catholic numerical dominance, as proposed by the MDH. Our tie-level data can establish whether mixing patterns exceed what structural and demographic explanations predict, but cannot directly observe salience or preference as psychological states.

2 Methods and Data

2.1 Data Source and Sample Design

We evaluate the dynamics of religious homophily and friendship choice using longitudinal social network and survey data from the *NetHealth Study* at the University of Notre Dame, a multi-year, NIH-funded data-collection project. *NetHealth* tracked a single cohort of undergraduate students from their entry as freshmen in Autumn 2015 through their graduation in Spring 2019. Initial recruitment enlisted around 700 students (yielding a total sample of $N = 722$ participants who took at least one survey), representing roughly 36% of the typical incoming freshman cohort. While this is an exceptionally high coverage rate for an intensive longitudinal ego-network study, the multi-tiered recruitment strategy (which included snowball sampling) may introduce selection bias into homophily estimates if students systematically recruited highly similar alters into the study. The project tracked their social networks, behaviors, and attributes over 8 waves of data collection fielded approximately every 3–4 months.

In each of the eight waves, respondents completed a name-generator survey with the prompt: "We would like to know who you consider to be in your social network. In the spaces below, please list up to 20 people with whom you spend time communicating or interacting." Beginning with Wave 3, respondents were presented with their previously nominated alters and were allowed to add up to 5 more, for a maximum of 25 alters. These name generators, coupled with name interpreters detailing each alter's characteristics and the nature of the relationship, yielded a unique multiwave,

long-term ego-network dataset that constitutes our primary data source. For the purposes of evaluating within-cohort homophily, we restrict alter nominations to those designated as fellow on-campus students.

2.2 Variables

2.2.1 Ego-Level (Participant) Measures

Ego-level attributes are drawn from the *NetHealth* Basic Survey, which captured demographic and behavioral characteristics across the eight waves. Baseline traits were measured at Wave 1. The primary independent variable is baseline *ego religion*. Respondents were categorized into four groups: Catholic (72.7%, the dominant majority), No Religion (12.3%), Protestant (9.8%), and Other Religion (4.6%). Missing responses account for $< 1\%$. Additional demographic controls include *ego gender*, coded as Woman (51.1%) or Man (47.9%), and *ego race/ethnicity*, categorized into White (65.0%), Latino/a (12.3%), Asian-American (8.7%), Foreign Student (7.1%), African-American (5.7%), and Other ($< 1\%$). To control for individual-level *religious salience (frequency of discussion)* independent of homophily, we use a measure of how frequently the respondent discusses religion (captured in Wave 2). Responses are recorded on a 5-point Likert scale ranging from "Not at all" (4.6%), "Less than 1-2 times a month" (11.8%), "1-2 times a month" (31.9%), "1-2 times a week" (27.4%), to "Three times a week or more" (12.6%).

2.2.2 Alter-Level (Tie) Measures

Alter characteristics and relationship contexts are derived from the *NetHealth* Network Survey, in which egos provided information about their nominated peers. For the *alter relationship context*, denoting the relation of the alter to the university (e.g., Student, Alum, Prospect), 48.6% of overall nominations were for fellow students. We subset our dyadic data to these alters ($N = 17{,}452$ student nominations), then drop 312 on-campus family nominations (e.g., siblings who both attend the university), leaving $N = 17{,}140$ student ties across Waves 1–8 (the regression analyses use Waves 3–8). Egos reported the perceived *alter religion* of each nominated fellow student (Catholic: 67.4%; Protestant: 6.9%; No Religion: 7.2%; Other Religion: 2.3%; Missing: 16.2%). This is used to construct our primary outcome variable, a binary indicator of religious homophily (coded 1 if ego and alter share the same religious affiliation, and 0 otherwise). Egos also reported each student *alter gender and race* (e.g., Woman: 52.2%; Man: 47.8%; and White: 77.8%, with 7.5% missing). These allow us to construct binary dyadic indicators for gender homophily and racial/ethnic homophily. To adjust for structurally induced clustering, egos indicated whether each fellow student shared *physical and social foci (roommates and dorms)*, noting whether the alter was a roommate (13.7% yes) and whether they resided in the same dormitory (35.3% yes); each item is missing for 31.2% of ties.[1] Finally, egos were asked to report how close they felt to each nominated alter on an ordered scale of *friendship closeness (intimacy)*. We construct an ordered numeric indicator for friendship intensity (coded 1 for "Distant", 2 for "Less Than Close", 3 for "Merely Close", and 4 for "Especially Close"). This continuous indicator is added to our multivariate models to control for baseline differences in the homophily of strong versus weak ties.

2.3 Opportunity Offset Formulation

To model the likelihood that a tie is homophilous on religion while adjusting for baseline group proportions, we employ a dyadic logistic regression with an opportunity offset (Scott and Wild,

[1] At Notre Dame, the dormitory system constitutes a primary axis of residential and social life.

1991). In highly skewed settings like Notre Dame, where Catholics make up nearly three-quarters of the student body, a traditional regression approach struggles to separate active friendship preferences from simple numeric availability. A Catholic student is highly likely to befriend another Catholic simply because the vast majority of their available peers are Catholic (Blau, 1977b), not necessarily because they are actively seeking out a same-religion friend. The opportunity offset corrects for this by subtracting the baseline probability of meeting someone of the same religion under conditions of random mixing.

Formally, we specify the model as:

$$\text{logit}(P(\text{Same Religion} = 1)) = \beta_0 + \beta_1 \text{EgoReligion} + \mathbf{X}\beta + \text{opportunity_offset} \tag{1}$$

In this equation, the opportunity_offset is defined as $\log\left(\frac{p_w}{1-p_w}\right)$, where p_w represents the overall proportion of the respondent's religious group in the available pool of peers for that specific wave. By forcing this term into the model with a fixed coefficient of 1.0, we adjust the baseline odds of tie formation. Consequently, the remaining coefficients represent the *excess* tendency to form a homophilous tie above and beyond what the structural opportunity pool dictates.

The model also includes $\mathbf{X}$, a vector of dyadic controls representing alternative mechanisms that might induce homophily without active religious sorting. These include demographic similarities (same gender and same race) as well as physical and social foci (being roommates or living in the same dorm). This specification provides a direct test of the MDH and MEH predictions. A positive β_1 coefficient for a minority group indicates that the group exhibits significantly stronger religious homophily than the Catholic majority, even after accounting for their smaller population size and alternative sorting mechanisms. Such a result would directly support the MDH, demonstrating that numeric minorities actively forge tighter in-group bonds in response to a dominant-majority environment.

Appendix A details the methodological advantages of this dyadic approach relative to two natural alternatives: aggregated E-I index measures, and mixed-effects multilevel models.

3 Results

3.1 Preliminary Checks

3.1.1 Opportunity Structure Stability

Before analyzing active homophily, it is crucial to establish the stability of the baseline opportunity structure. If the numeric composition of the student body fluctuated wildly over the four years, any changes in tie formation could simply be an artifact of changing demographics rather than evolving social preferences. The wave-specific proportions of available student peers across Waves 3 to 8 are plotted in Figure 1. The wave-specific shares are nearly constant: Catholic students make up 77–82% of the alter pool in each wave, No Religion 7–11%, Protestant 8–10%, and Other Religion 2–3%. This stability makes it unlikely that changes in the homophily estimates in subsequent models reflect demographic drift rather than behavioral sorting.

3.1.2 Baseline Mixing Patterns (Yule's Q)

To address the group-size artifact, we first evaluate the longitudinal trajectory of religious mixing using group-level, baseline-adjusted Yule's Q across all eight waves of the college cohort (see Figure 2). Unlike alternative aggregate metrics—such as raw in-group tie percentages, Coleman's

homophily index, or Krackhardt and Stern's E-I (External-Internal) index—Yule's Q is independent of relative group sizes (Borgatti et al., 2024, 159). Because traditional measures like the E-I index are strictly bounded by the available opportunity pool, they inherently force numeric minorities to exhibit higher rates of heterophily simply because of their small baseline population, thereby distorting cross-group comparisons (for a review mapping segregation measures to theoretical properties, see Bojanowski and Corten, 2014).

In contrast, Yule's Q is a direct transformation of the odds ratio, rendering it margin-free: unlike the E-I index, its expected value under random mixing is 0 regardless of group size. This makes Yule's Q well suited to testing, *within each group*, whether observed mixing departs from what that group's own random-mixing baseline would predict. Margin-invariance of the expected value does not, however, imply that raw Q magnitudes are comparable *across* groups of very different sizes. A degree-preserving permutation analysis (resampling alters within each wave's observed opportunity pool while holding each ego's degree fixed) shows that, for groups as small as Other Religion, the null distribution of Q is itself both wide and biased away from zero, so a given raw Q is far less surprising for a small group than the identical value would be for a large one. We therefore use Yule's Q to establish that each group, individually, sorts more than its own chance baseline predicts, rather than to rank the relative strength of homophily across groups of unequal size.

Yule's Q is defined as:

$$Q = \frac{ad - bc}{ad + bc} \tag{2}$$

Where a is the number of ingroup ties (e.g., Catholics befriending Catholics), b is the number of outgroup ties from group G (Catholics befriending non-Catholics), c is the number of ties from the outgroup to group G (non-Catholics befriending Catholics), and d is the number of ties strictly between outgroup members (non-Catholics befriending non-Catholics).

The resulting index ranges from -1 to $+1$. A value of 0 indicates perfect random mixing, in which ties form without regard to religion. A positive value ($Q > 0$) indicates "inbreeding homophily" (Marsden, 1988): an excess of same-group ties relative to what random mixing would predict, consistent with active sorting toward same-group friends. At the extreme, a Q of $+1$ implies absolute homophily, where group members only ever befriend their own kind. Conversely, a negative value ($Q < 0$) indicates heterophily: an excess of cross-group ties relative to chance, consistent with active avoidance of same-group ties, with -1 indicating that every friendship crosses religious lines.

Yule's Q is positive for every religious group in every wave, which is consistent with *universal inbreeding homophily* ($Q > 0$): over the four years of college, each group's same-religion friendships exceed the rate expected under its own random-mixing baseline in descriptive terms. The permutation test below indicates that this departure is statistically significant in all but one group-wave.

Rather than relying on the raw magnitude of Q alone, we anchor this claim to a formal permutation test against each group's chance baseline, presented alongside our other cross-group precision diagnostics in Section 3.3.

3.2 Primary Analysis: Pooled Longitudinal Regressions with Robust Clustered Standard Errors

Our primary test of the MDH against the MEH pools ties across all six waves for which the full covariate battery is available (Waves 3–8), rather than resting on any single cross-section: this maximizes statistical power, lets each group's coefficient average over wave-specific idiosyncrasies,

and is the specification we treat as decisive for adjudicating between the two hypotheses. We fit a nested series of pooled models and an interaction model (Table 1). Because individual students are observed repeatedly, we estimate robust standard errors clustered at the respondent level. The pooled data show all three non-Catholic groups retaining a positive, statistically significant effect on active homophily in each of Models 1–3: No Religion ($\beta = 0.556, p < 0.01$, Model 2; $\beta = 0.550, p < 0.01$, Model 3), Protestant ($\beta = 0.496, p < 0.01$), and Other Religion ($\beta = 1.718$–$1.727, p < 0.01$), while the Catholic baseline (the Model 1 intercept, i.e., the excess log-odds of same-religion ties among Catholic students relative to the opportunity-implied rate) is small but reliably positive ($\beta = 0.267$, $p < 0.001$).

Furthermore, note that we find a negative effect of same-dorm co-residence on the probability of forming a same-religion tie ($\beta = -0.249, p < 0.05$, Model 2), which weakens to a marginally significant effect once ego/alter degree controls are added ($\beta = -0.245, p \approx 0.053$, Model 3). This is consistent with the institutional policy of diversifying dorms by religious affiliation when students are assigned, based on data obtained at matriculation, which means that dorms, as social foci (Feld and Grofman, 2009), expose students to others of different religions. Dorms are particularly powerful foci in this context because, unless exceptional circumstances arise, students remain in the (same-gender) dorm they are assigned to at admission for the entirety of their college experience. In contrast to dorms, race homophily (the dyadic tendency to bond with same-race peers) is a statistically significant *positive* predictor of same-religion tie formation ($\beta = 0.235, p < 0.05$, Model 2; weakening to a marginal effect, $\beta = 0.227, p \approx 0.053$, once degree controls are added in Model 3), indicating that religious and racial boundaries operate as complementary, rather than competing, axes of social sorting—a classic case of Blau's principle of Multiform Heterogeneity, whereby making a tie based on one kind of similarity leads to making a tie along another type of similarity.[2]

The results show that *non-Catholic groups collectively maintain elevated, opportunity-adjusted homophily relative to the Catholic majority, which does not change appreciably over time*, as evidenced by broadly parallel trends across waves. We caution against reading the specific magnitudes across Protestant, No Religion, and Other Religion as a stable ranking; Section 3.3 presents a pooled bootstrap sensitivity analysis that directly substantiates this caution, in the same spirit as the Yule's Q permutation test above.

To evaluate the group differences on the probability scale within a non-linear logit framework (Mize, 2019), we compute average marginal effects of each non-Catholic affiliation relative to the Catholic baseline. Because raw probabilities are confounded by unequal group sizes (the group-size artifact), we set the opportunity offset to 0 (equivalent to a 50% share for every group) and average the predicted contrast over all observed ties in each wave. This isolates the excess probability of an active homophilous tie under an equal-opportunity counterfactual. Figure 3 plots these wave-specific average marginal effects, each estimated from a separate cross-sectional opportunity-adjusted logistic regression that includes the Model 2 covariates except religious salience. The Other Religion estimate is positive in every wave (roughly 0.28–0.36, with confidence intervals excluding zero). The No Religion and Protestant estimates are positive in most waves (roughly 0.04–0.19), but their intervals include zero in Waves 3 and 7 (No Religion) and Wave 6 (Protestant). The relative ordering among the three non-Catholic groups should not be read as a reliable ranking, given the small and uneven sample sizes underlying these estimates.

[2] Adding each ego's log out-degree and each alter's log in-degree (Model 3) leaves the three group coefficients essentially unchanged, indicating that opportunity-adjusted homophily is not an artifact of differential nomination activity or popularity by religious group. Ego activity is itself a marginally significant positive predictor ($p \approx 0.04$–0.05); alter popularity is not ($p \approx 0.13$–0.16). Adding the degree controls also pushes the dorm and race-homophily effects just past the conventional $p < 0.05$ threshold (to $p \approx 0.053$ for both), so those two structural effects should be read as suggestive rather than definitive.

3.3 Robustness Checks: The Group-Size Precision Problem

The descriptive Yule's Q trajectories and the pooled regression coefficients above share a common vulnerability, noted at each point above: a statistic can be correctly bias-adjusted for group size and still be far noisier for a small group than for a large one. Bias correction and precision are separate properties, and only the former is guaranteed by the opportunity offset or by Yule's Q's margin-invariance. We address this concern with two independent diagnostics—one for the descriptive Yule's Q trajectories, one for the pooled regression coefficients—that converge on the same conclusion.

3.3.1 Yule's Q Permutation Test

Rather than relying on the raw magnitude of Q alone, we anchor the universal-inbreeding-homophily claim to a degree-preserving permutation test: for each ego, we hold degree and own religion fixed and redraw nominated alters' religion from that wave's observed opportunity pool, generating a null distribution of Q against which the observed value is standardized. As Table 2 shows, every group's observed Q lies above its own permutation null mean in every wave (z-scores range from 1.3 to 7.5), and the difference is significant at $p < 0.05$ in all but one group-wave combination (Protestant, Wave 1, $p = 0.086$). This test speaks only to whether each group, individually, departs from its own chance baseline; it is not a basis for ranking groups against one another, since a group's null variance itself scales with its size.

3.3.2 Bootstrap Sensitivity Analysis of Pooled Coefficients

A parallel concern applies to the pooled regression coefficients: does the apparent ordering of point estimates (Other Religion > Protestant ≈ No Religion) reflect a real ranking, or merely the wider sampling variability of smaller groups? A pooled bootstrap analysis across Waves 3–8 (resampling the 472 unique egos who contribute ties, with replacement, separately within each religious group, so that all of a resampled ego's ties across waves move together as a single block; full results in Table 3) confirms that Protestant and No Religion students each show a modest but statistically robust excess in-group tie rate relative to their own chance baseline (bootstrap mean excess ≈ 9 and 14 percentage points, respectively, both with 95% CIs excluding zero), and that this excess is reliably larger than the Catholic majority's own small excess (≈ 4 percentage points). Other Religion's point estimate is comparably large (≈ 13 percentage points), but—limited to only 22 unique individuals across the six-wave panel in this baseline sample (19 in the fully adjusted sample of Table 1)—its 95% CI is wide enough that it remains statistically indistinguishable from every other group, including the Catholic majority. This indistinguishability persists after pooling six waves of data because the binding constraint is the number of distinct people in the group, not the number of observed ties: additional waves largely add repeated observations from the same small set of individuals rather than new independent information.

As a complementary check on the same question, we also computed Wald tests of pairwise equality directly on the pooled log-odds coefficients (using the same ego-clustered variance structure as the pooled regression tables), rather than on the baseline-adjusted probability-scale excess rates used above. We test all six pairwise contrasts among the four groups, Holm-corrected for multiple comparisons (Table 3, Panels C–D): the three Catholic-versus-non-Catholic contrasts, included as a sanity check on the distinction the rest of this analysis treats as reliable, and the three non-Catholic-versus-non-Catholic contrasts of substantive interest. On the baseline Model 1 specification (Panel C), every Catholic-versus-non-Catholic contrast is significant after correction (all Holm-adjusted $p < 0.03$), confirming that sanity check. Two non-Catholic contrasts are nominally significant before

correction (No Religion versus Other Religion, $p = 0.038$; Other Religion versus Protestant, $p = 0.029$), but neither survives Holm correction (both $p = 0.086$), and No Religion versus Protestant is clearly null ($p = 0.84$). Re-running the identical six contrasts on the fully-adjusted Model 3 specification (Panel D, structural foci, demographic homophily, salience, and degree controls) leaves this conclusion unchanged—no non-Catholic pairwise contrast survives Holm correction (all Holm-adjusted $p > 0.08$). The raw p-values for the same two contrasts are similar on Model 3 ($p = 0.033$ and $p = 0.027$), so the pattern does not change once structural, salience, and degree covariates are added. This log-odds-scale test agrees with the probability-scale bootstrap comparisons above.

Together, these three independent diagnostics—testing each group's Yule's Q against its own permutation null, testing pairwise differences in bootstrap-resampled excess tie rates, and testing pairwise equality of the pooled log-odds coefficients directly—converge on the same conclusion: the data reliably support treating Catholic and non-Catholic as distinct categories in this population, but they do not reliably support a finer ranking among Protestant, No Religion, and Other Religion students.

4 Discussion and Concluding Remarks

Our results support the MDH relative to the MEH predictions when religious groups are considered at the Catholic/non-Catholic level, but not for a finer-grained ranking among non-Catholic subgroups. The baseline homophily of the Catholic majority is largely structural—a byproduct of their sheer numeric availability within the university environment, consistent with the more streamlined version of Blau's (1977b) macro-structural theory: its opportunity-adjusted excess is small (a bootstrap mean of about 4 percentage points, Table 3, Panel A), although reliably above zero. Because Catholics constitute nearly three-quarters of the student body, random mixing naturally yields highly homophilous networks.

In contrast, non-Catholic students collectively exhibit opportunity-adjusted homophily above chance. Surrounded by a dominant majority, their minority identity may become more salient, prompting more active maintenance of social boundaries around subcultural beliefs and practices. This holds for No Religion as well as for Protestant and Other Religion students: the No Religion coefficient is positive and statistically significant in each of Models 1–3 of Table 1, including after adjusting for structural controls and for ego/alter degree ($\beta = 0.549$ to 0.556, $p < 0.01$). In the interaction specification, which uses Wave 3 as the reference and adds group-by-wave terms, the No Religion main effect is smaller and not statistically significant ($\beta = 0.370$, $p \approx 0.11$), while the Other Religion and Protestant effects remain significant. However, as detailed above, our data cannot reliably differentiate whether Protestants, students with no religion, or students of other religions exhibit a stronger version of this tendency than one another: the Other Religion group's large point estimate reflects a denominator of only 22 unique individuals pooled across all six waves, and a bootstrap sensitivity analysis finds no statistically reliable differences among the three non-Catholic groups. We therefore interpret the evidence as supporting a Catholic/non-Catholic distinction in active homophily rather than a specific ordering among minority religious groups.

Furthermore, the wave-specific marginal effects in Figure 3 suggest that the elevated rate of active homophily among non-Catholic students, considered collectively, does not appear to diminish over the four years of college. The group-by-wave interaction terms in Table 1 are not statistically significant for any group. Given the small number of unique egos in the non-Catholic groups, these tests have limited power, so this is absence of evidence for change rather than evidence of stability. These groups maintain broadly parallel and stable trajectories of in-group affinity relative to the Catholic baseline, despite the potential integrating forces of shared university spaces and

randomized residential dorm assignments. As noted above, however, the data cannot support claims about which specific non-Catholic group maintains the most durable boundary.

Finally, we note several limitations of the current study. First, there is a substantial selection effect inherent to our sample: non-Catholic students have actively chosen to attend a Catholic university. As a result, they may be less culturally distinct from the Catholic majority than they would be in a secular setting, which may attenuate the boundaries we observe and shift the baseline mechanics of out-group exclusion. Second, because alter attributes are ego-reported, measurement error may be present. In a setting where Catholicism is dominant, respondents who do not know a friend's religion might assume that they are Catholic, potentially artificially inflating Catholic homophily or cross-group ties toward Catholics. Third, we observe a single institutional context with one specific distribution of religious groups. As previous research has shown (Goodreau et al., 2009; McFarland et al., 2014), the strength of homophily can vary contextually, often exhibiting curvilinear relationships with group proportions rather than strictly linear responses to minority or majority status. Future work extending these findings across multiple institutional contexts could yield greater theoretical specificity regarding the separable effects of group size and group proportion.

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A Methodological Advantages of the Dyadic Approach

A.1 Benefits of the Opportunity Offset

The Krackhardt and Stern (1988) External-Internal (E-I) index is defined as $EI = (E-I)/(E+I)$. Under random mixing, the expected E-I index for group G is a direct linear function of group size p:

$$EI_{\text{expected}} = (1-p) - p = 1 - 2p \tag{3}$$

Because the baseline shifts with p, raw E-I indices cannot be compared across groups of different sizes to infer preferences. Individual-level aggregations of normalized E-I indices also introduce a severe boundary bias: any minority ego with zero same-religion ties is assigned an EI_{norm} of exactly +1.0. Averaging these individual scores heavily pulls the minority group's average toward +1.0, falsely indicating heterophily. Our dyadic logistic regression with an opportunity offset ($\log(p/(1-p))$) mitigates these biases. Entering this offset with its coefficient constrained to exactly 1.0 subtracts the baseline opportunity structure. Consequently, the model intercept and group coefficients represent the *excess log-odds of homophilous tie formation beyond what is expected under random mixing*, providing a control-integrated point estimate that is not subject to the E-I index's boundary and aggregation biases. This addresses bias in the point estimate itself; it does not guarantee that coefficients are equally *precise* across groups of very different sizes. As our Results show, groups with few unique egos (e.g., Other Religion) yield coefficients with substantially wider uncertainty, which limits how confidently their magnitude can be compared to other groups' even though the point estimates themselves are not systematically biased by group size.

A.2 Why Dyadic Clustered Regression is Preferable to Mixed-Effects Multilevel Models

In social network analysis, egocentric network datasets are inherently hierarchical: nominated friendship ties (Level-1) are nested within individual egos (Level-2 clusters). To correct for this nesting, methodologists frequently recommend mixed-effects logistic regressions:

$$\log\left(\frac{P(Y_{ij}=1 \mid \gamma_i)}{1-P(Y_{ij}=1 \mid \gamma_i)}\right) = \beta_0^C + \beta_1^C \text{EgoReligion}_i + \beta_2^C \mathbf{X}_{ij} + \gamma_i + \text{offset}_{wi} \tag{4}$$

Where $\gamma_i \sim N(0, \sigma^2)$ is a random intercept for ego i, suggesting that multilevel models can be a useful way to model this type of data (Snijders and Baerveldt, 2003). But it can be shown that, for the purposes of answering our questions, the mixed-effects multilevel model is statistically and conceptually less appropriate than a population-average dyadic logistic regression with robust clustered standard errors. First, in a mixed-effects logistic model, the estimated fixed effects (β^C) are conditional: they represent the change in log-odds for a *specific individual* if that individual's religion were to change. But if religion is a relatively stable baseline trait, as it is in our data, then this is a somewhat less-than-realistic modeling assumption. In contrast, our model with robust clustered standard errors represents a marginal (population-average) model:

$$\log\left(\frac{P(Y_{ij}=1)}{1-P(Y_{ij}=1)}\right) = \beta_0^M + \beta_1^M \text{EgoReligion}_i + \beta_2^M \mathbf{X}_{ij} + \text{offset}_{wi} \tag{5}$$

Substantively, because our core hypothesis concerns group-level differences across the population, the marginal population-average model is the more appropriate approach.

Secondly, the conditional parameters estimated by mixed-effects models are always larger in absolute value than the marginal parameters, indicating that the unobserved ego-level heterogeneity is large and that the random intercepts γ_i act as a "sponge," absorbing the unobserved cluster-level variance. This collinearity causes the fixed effects in the MLM to shrink toward zero, stripping the model of its statistical power.

Finally, because Protestant and Other Religion students constitute relatively small percentages of the student body, the probability that a minority ego has *exactly zero* same-religion friends is high. When an ego has zero same-religion friends, their Level-1 outcome Y_{ij} has zero variance. In a mixed-effects logistic regression, the maximum likelihood estimate of the random intercept for an ego with zero successes is negative infinity ($\hat{\gamma}_i \to -\infty$). When the numerical solver attempts to estimate the random effects for these zero-variance clusters, the algorithm fails to find stable starting values and diverges. Our dyadic logistic regression with robust clustered standard errors avoids these numerical and conceptual pitfalls: it is numerically stable, conceptually accurate, and produces statistically defensible standard errors.

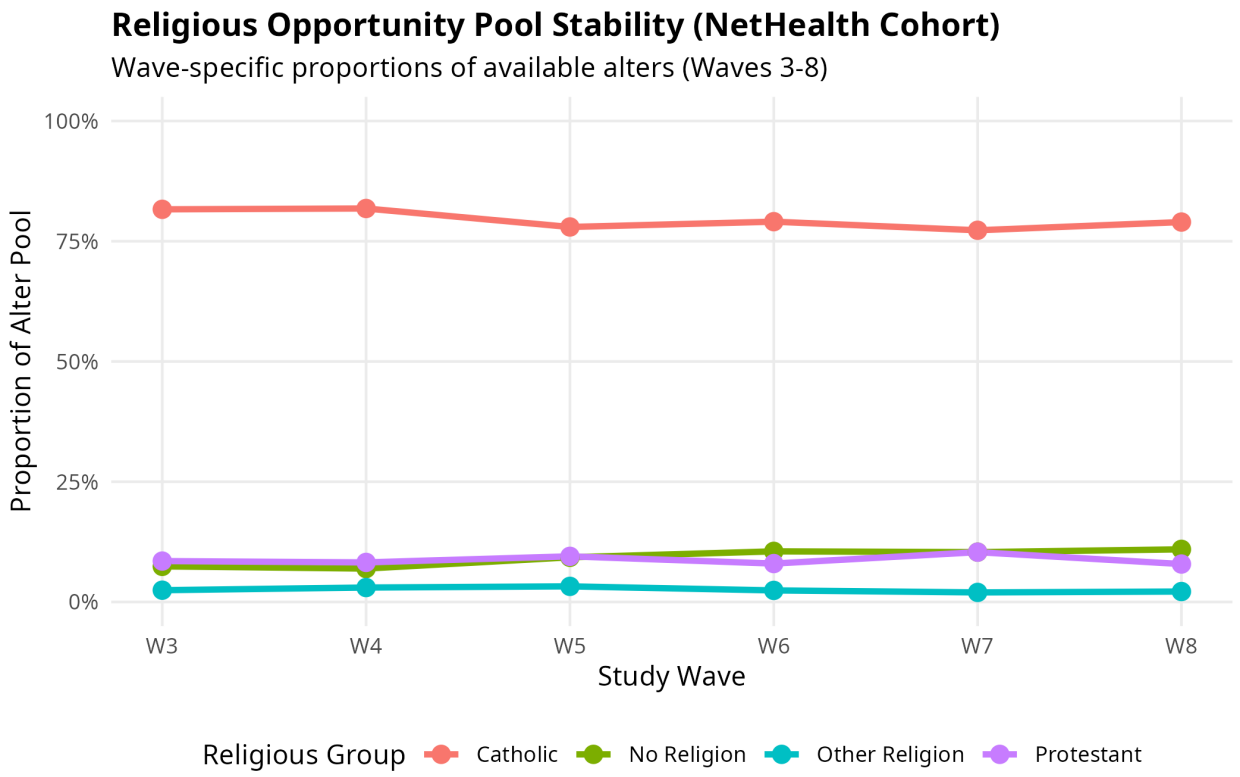


Figure 1: Preliminary Check: Religious Opportunity Structure Stability (Proportions of Available Peers, Waves 3-8).

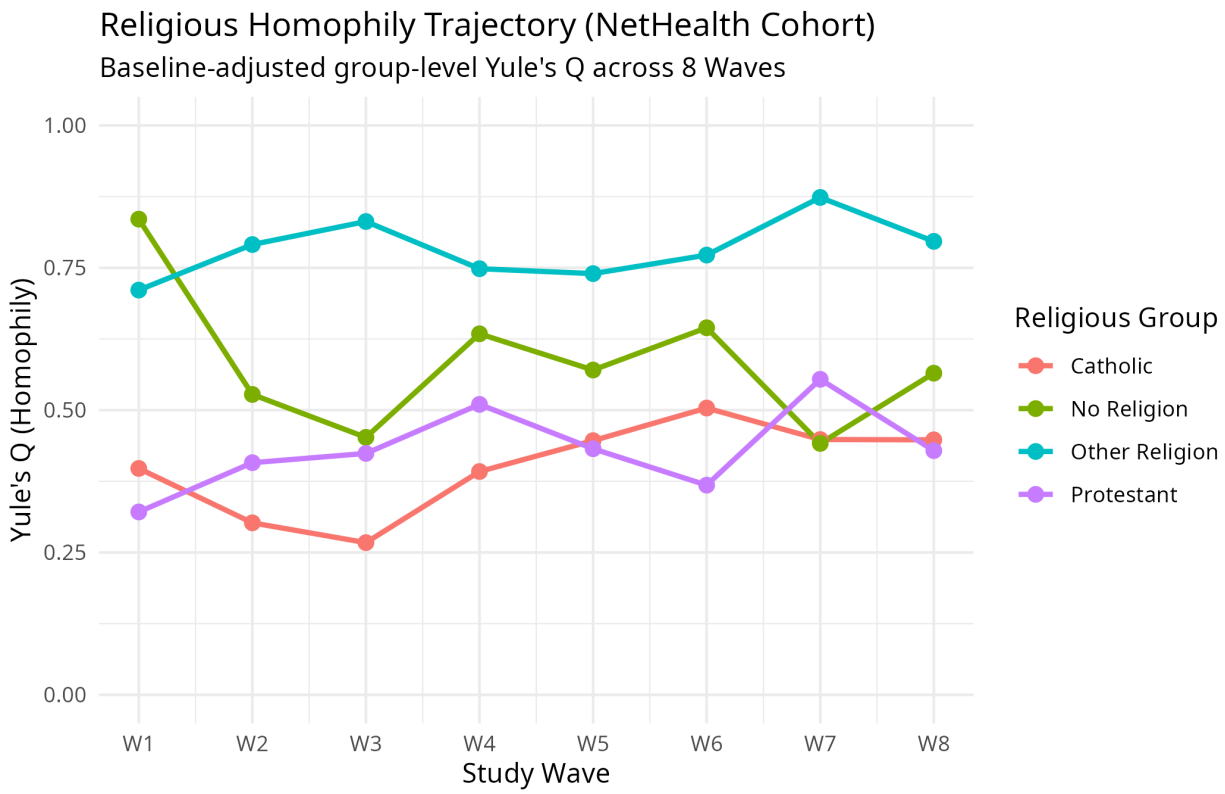


Figure 2: Preliminary Check: Longitudinal Trajectories of Religious Homophily (Yule's Q) across Waves 1 to 8.

Table 1: Primary Analysis: Pooled Longitudinal and Interaction Models (Waves 3-8, Clustered SEs).

	Model 1	Model 2	Model 3 (+Degree)	Interaction (+Degree)
Intercept	0.267*** (0.065)	0.248 (0.315)	-0.120 (0.421)	-0.163 (0.441)
Ego: No Religion	0.549** (0.197)	0.556** (0.196)	0.550** (0.195)	0.370 (0.234)
Ego: Other Religion	1.718** (0.536)	1.727*** (0.522)	1.718** (0.524)	1.741** (0.560)
Ego: Protestant	0.497** (0.182)	0.496** (0.184)	0.496** (0.185)	0.612** (0.227)
Tie: Gender Homophily	—	0.141 (0.138)	0.134 (0.137)	0.131 (0.137)
Tie: Race Homophily	—	0.235* (0.119)	0.227 (0.118)	0.225 (0.118)
Tie: Roommates	—	-0.131 (0.108)	-0.097 (0.109)	-0.098 (0.109)
Tie: Same Dorm	—	-0.249* (0.127)	-0.245 (0.127)	-0.240 (0.127)
Tie: Subjective Closeness	—	-0.057 (0.076)	-0.039 (0.076)	-0.041 (0.076)
Ego: Religious Salience	—	0.032 (0.060)	0.034 (0.060)	0.032 (0.060)
Ego: Activity (log out-deg.)	—	—	0.187* (0.094)	0.191* (0.094)
Alter: Popularity (log in-deg.)	—	—	-0.191 (0.126)	-0.178 (0.128)
Tie: Wave Number	—	—	—	0.016 (0.027)
Ego: No Religion x Wave	—	—	—	0.076 (0.080)
Ego: Other Religion x Wave	—	—	—	-0.010 (0.110)
Ego: Protestant x Wave	—	—	—	-0.053 (0.072)
Group sample sizes (unique egos / dyadic ties), Waves 3–8 pooled, fully-adjusted sample				
Catholic	326 / 7,393			
No Religion	37 / 670			
Other Religion	19 / 348			
Protestant	46 / 1,111			

Note: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$

Egos are counted once even if they contribute ties across multiple waves; standard errors clustered by egoid throughout. On-campus family ties (e.g., siblings both enrolled and nominating each other) are excluded from the analytic sample throughout, so all reported ties are nominations coded as student (non-family) relationships. Model 3 and the Interaction model add each ego's log out-degree (nominations made that wave) and each alter's log in-degree (nominations received that wave), computed from the full wave-specific nomination roster, to test whether opportunity-adjusted homophily reflects a degree/activity confound rather than affiliation itself (Reviewer 1). Neither degree term meaningfully attenuates the three group coefficients, though ego activity is itself marginally associated with same-religion tie formation ($p \approx 0.04$–0.05). The Other Religion group's small N (19 egos in this analytic sample, across the six-wave panel) implies substantially lower precision for its coefficient than for the other groups; see Results for a bootstrap sensitivity analysis.

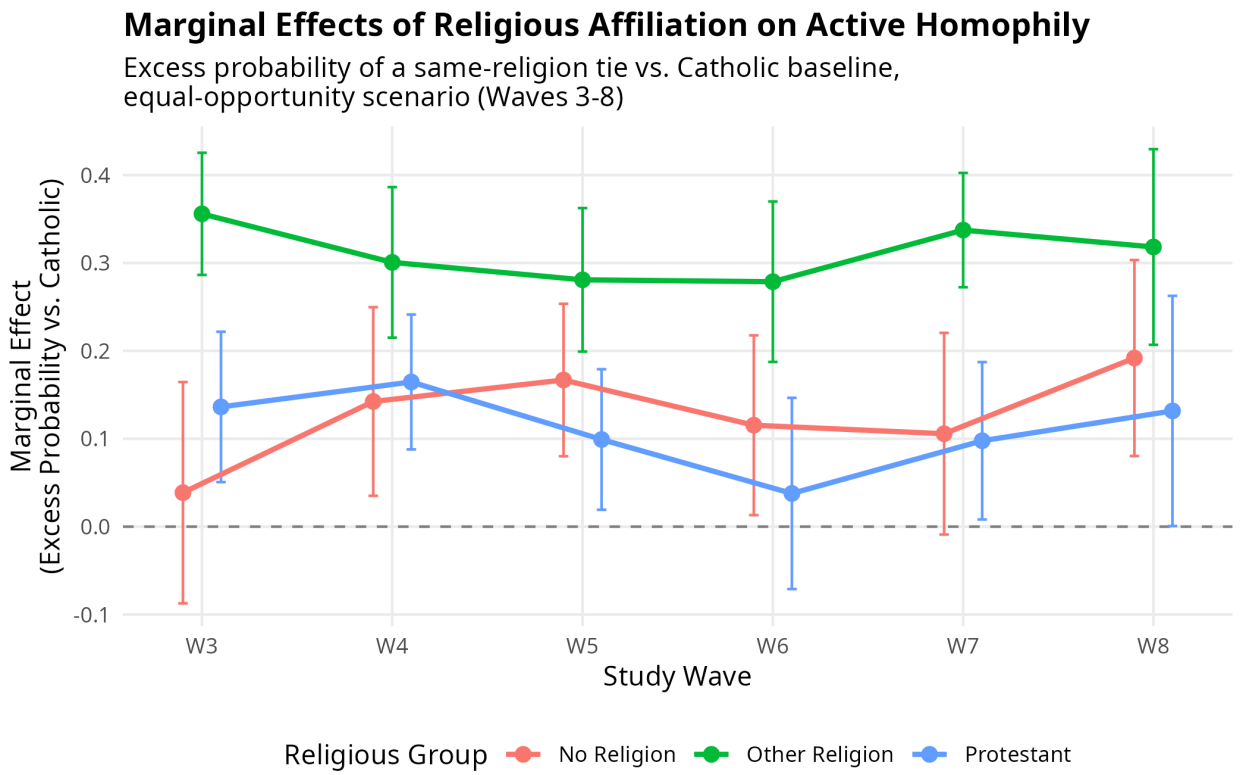


Figure 3: Primary Analysis: Average Marginal Effects of Religious Affiliation on Active Homophily over Time (Waves 3-8). Each point is the wave-specific average marginal effect, on the probability scale, of belonging to the given non-Catholic group rather than Catholic. Each wave is a separate cross-sectional logistic regression with the Model 2 covariates except religious salience; the effect is averaged over all observed ties in that wave, with the opportunity offset fixed at 0 (an equal-opportunity counterfactual); error bars show 95% confidence intervals.

Table 2: Robustness Check: Degree-Preserving Permutation Test of Yule's Q (Waves 1–8).

Wave	Group	Observed Q	Null Mean	Null SD	z	p
W1	Catholic	0.398	-0.001	0.123	3.23	0.001
	No Religion	0.835	-0.042	0.259	3.39	<0.001
	Other Religion	0.711	-0.373	0.668	1.62	0.015
	Protestant	0.321	-0.068	0.310	1.26	0.086
W2	Catholic	0.302	0.001	0.049	6.14	<0.001
	No Religion	0.528	0.001	0.098	5.37	<0.001
	Other Religion	0.791	-0.107	0.366	2.45	<0.001
	Protestant	0.408	-0.005	0.104	3.96	<0.001
W3	Catholic	0.267	0.001	0.067	3.96	<0.001
	No Religion	0.452	-0.008	0.147	3.12	<0.001
	Other Religion	0.831	-0.197	0.514	2.00	<0.001
	Protestant	0.424	-0.007	0.128	3.37	<0.001
W4	Catholic	0.392	-0.003	0.069	5.74	<0.001
	No Religion	0.634	-0.022	0.158	4.15	<0.001
	Other Religion	0.748	-0.111	0.374	2.30	<0.001
	Protestant	0.510	-0.010	0.128	4.05	<0.001
W5	Catholic	0.446	-0.000	0.062	7.24	<0.001
	No Religion	0.570	-0.018	0.146	4.05	<0.001
	Other Religion	0.740	-0.098	0.372	2.26	<0.001
	Protestant	0.432	-0.005	0.118	3.71	<0.001
W6	Catholic	0.504	-0.003	0.068	7.46	<0.001
	No Religion	0.645	-0.017	0.142	4.66	<0.001
	Other Religion	0.772	-0.143	0.462	1.98	<0.001
	Protestant	0.368	-0.017	0.137	2.80	0.001
W7	Catholic	0.448	-0.004	0.075	6.04	<0.001
	No Religion	0.442	-0.016	0.172	2.65	<0.001
	Other Religion	0.874	-0.262	0.565	2.01	<0.001
	Protestant	0.554	-0.013	0.131	4.32	<0.001
W8	Catholic	0.448	-0.000	0.100	4.49	<0.001
	No Religion	0.565	-0.028	0.207	2.87	0.001
	Other Religion	0.796	-0.313	0.636	1.74	<0.001
	Protestant	0.429	-0.024	0.221	2.05	0.003

Note: Null distributions (1,000 permutations per wave) hold each ego's degree and own religion fixed and redraw nominated alters' religion from that wave's observed opportunity pool. Observed Q lies above its null mean in 32 of 32 group-waves. The standardized difference is significant at $p < 0.05$ in 31 of 32 group-waves; the exception(s): Protestant (Wave 1, $p = 0.086$). Because null variance itself scales with group size, z-scores should not be compared across groups to infer a ranking of homophily strength – differences partly reflect differential statistical power rather than differential preference. Cross-group comparisons are addressed separately via the bootstrap sensitivity analysis (Table 3).

Table 3: Robustness Check: Ego-Level Cluster Bootstrap Sensitivity Analysis of Excess In-Group Tie Rates (Waves 3–8, 2,000 replicates).

Panel A: Group-level excess in-group tie rate

Group	n Egos	n Ties	Observed Rate	Baseline Rate	Excess (95% CI)
Catholic	348	7,630	0.835	0.797	0.038 (0.021, 0.055)
No Religion	52	850	0.226	0.089	0.138 (0.083, 0.199)
Other Religion	22	381	0.155	0.026	0.125 (0.027, 0.275)
Protestant	50	1,185	0.181	0.088	0.094 (0.049, 0.146)

Note: Catholic's own excess (95% CI excludes zero) indicates that even the religious majority exhibits some in-group tie preference beyond pure chance mixing; the substantive question of interest is not whether any group departs from chance (all four do, per Table 2 and here), but the *relative magnitude* of that departure across groups, which Panel B addresses directly. The bootstrap uses the baseline sample (no covariate filtering), which is larger than the fully adjusted sample used for Table 1.

Panel B: Pairwise differences in excess rate

Group 1	Group 2	Mean Diff.	95% CI	Excludes 0
Catholic	No Religion	-0.100	(-0.163, -0.042)	Yes
Catholic	Other Religion	-0.087	(-0.239, 0.013)	No
Catholic	Protestant	-0.056	(-0.109, -0.007)	Yes
No Religion	Other Religion	0.013	(-0.150, 0.138)	No
No Religion	Protestant	0.044	(-0.030, 0.126)	No
Other Religion	Protestant	0.031	(-0.086, 0.192)	No

Note: Unique egos are resampled with replacement, separately within each religious group, so that all of a resampled ego's ties (across whatever waves they contribute to) move together as a single block. Excess rate is the observed same-religion tie rate minus the wave-specific opportunity/chance baseline used to construct the logit offset in the pooled dyadic regressions (Table 1); this baseline-adjusted quantity is comparable across groups of unequal size, unlike a raw tie rate or a log-odds coefficient. Panel B differences are valid because each group's bootstrap replicates are drawn independently of the other groups'.

Panel C: Wald tests of pairwise equality, pooled log-odds coefficients (Model 1, baseline)

Group 1	Group 2	Log-Odds Diff.	SE	χ^2	p	Holm p
Catholic	No Religion	0.549	0.197	7.766	0.005	0.027
Catholic	Other Religion	1.718	0.536	10.291	0.001	0.008
Catholic	Protestant	0.497	0.182	7.452	0.006	0.027
No Religion	Other Religion	-1.169	0.563	4.307	0.038	0.086
No Religion	Protestant	0.052	0.252	0.043	0.836	0.836
Other Religion	Protestant	1.221	0.558	4.786	0.029	0.086

Note: Wald tests of the equality of all six pairwise pooled Model 1 log-odds coefficients (opportunity-offset only, no structural/degree covariates; same fully adjusted analytic sample as the Model 1 column of Table 1), using the same ego-clustered variance-covariance matrix (clustered on `egoid`) as the pooled regression tables. The three Catholic-vs.-X contrasts are algebraically equivalent to testing each coefficient against zero and are included as a diagnostic sanity check, not new information. p-values are Holm-adjusted across all six contrasts. This is a complementary check to Panels A-B: it compares coefficients directly on the log-odds scale rather than baseline-adjusted probability-scale excess rates.

Panel D: Wald tests of pairwise equality, pooled log-odds coefficients (Model 3, fully adjusted)

Group 1	Group 2	Log-Odds Diff.	SE	χ^2	p	Holm p
Catholic	No Religion	0.550	0.195	7.974	0.005	0.024
Catholic	Other Religion	1.718	0.524	10.733	0.001	0.006
Catholic	Protestant	0.496	0.185	7.153	0.007	0.030
No Religion	Other Religion	-1.168	0.549	4.532	0.033	0.082
No Religion	Protestant	0.053	0.251	0.045	0.831	0.831
Other Religion	Protestant	1.222	0.553	4.877	0.027	0.082

Note: Identical contrasts to Panel C, but on the fully-adjusted Model 3 specification (Table 1: structural foci, demographic homophily, religious salience, and ego/alter log-degree controls), to check whether the pairwise conclusions are sensitive to adding these covariates. p-values are Holm-adjusted across all six contrasts.